

Synoptic CB: Macrophyte Fluxes

2025 Samples

2026-02-23

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#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2025_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("ch4_", .)
  )
```

```
co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2025_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("co2_", .)
  )
```

Combine all files

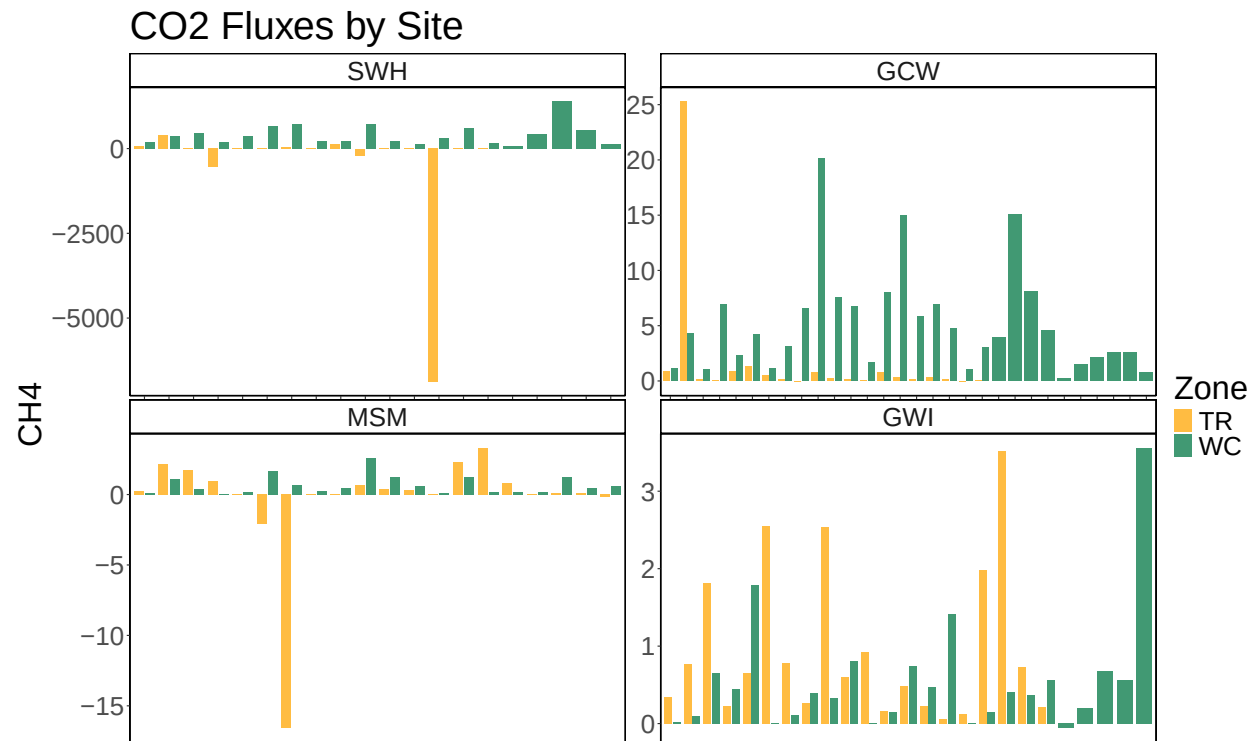
```
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_1"))
```

##Rename Zone

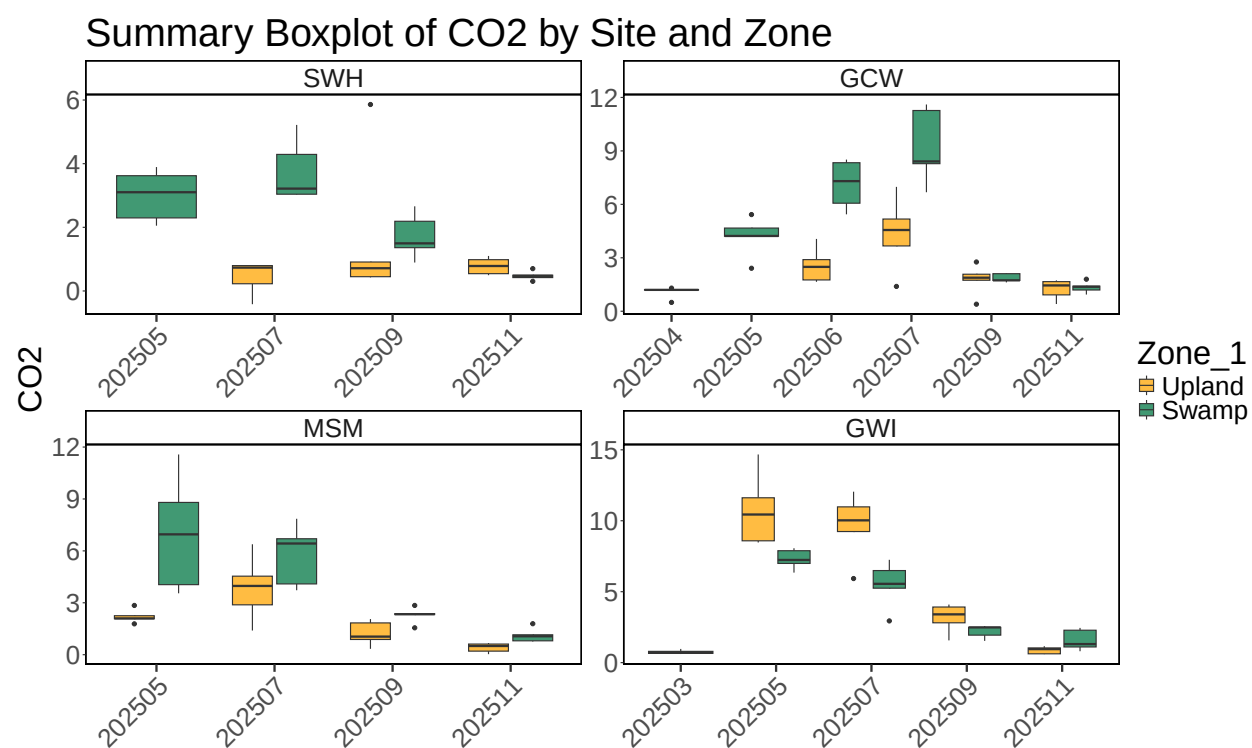
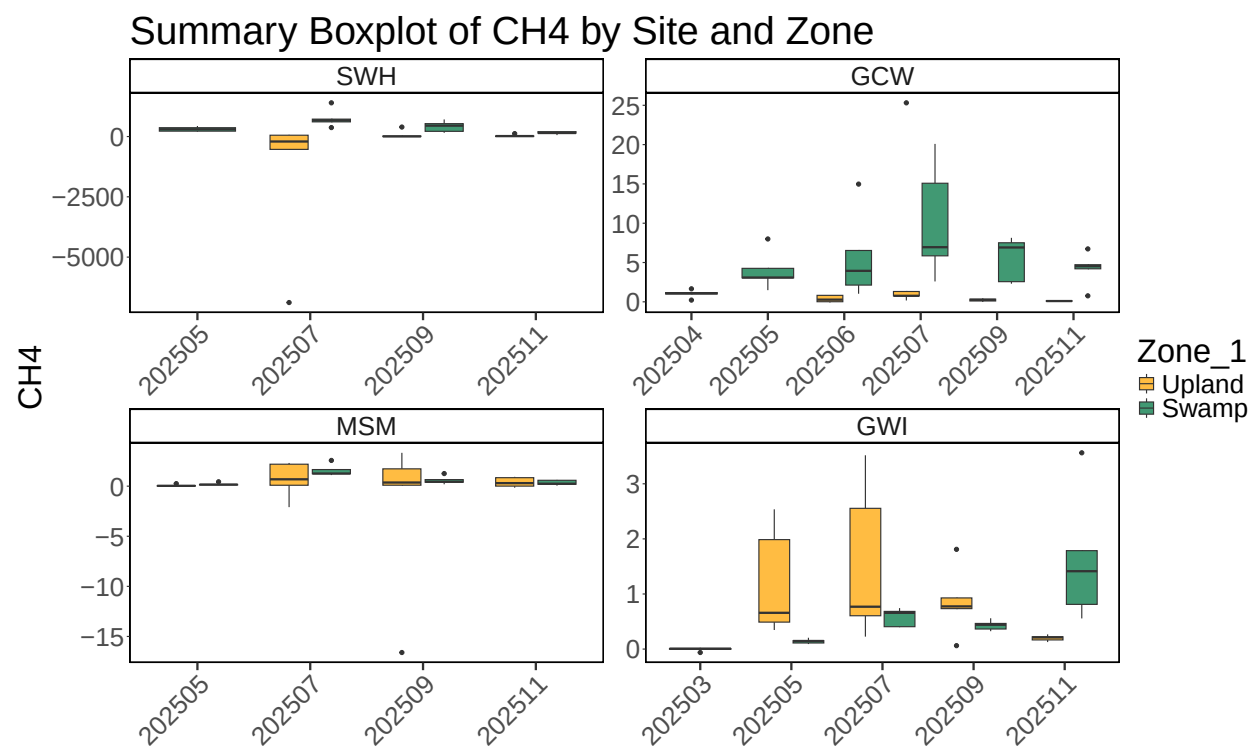
```
all_dat$Zone <- gsub("W", "WC", all_dat$Zone)
```

0.1 Write out files

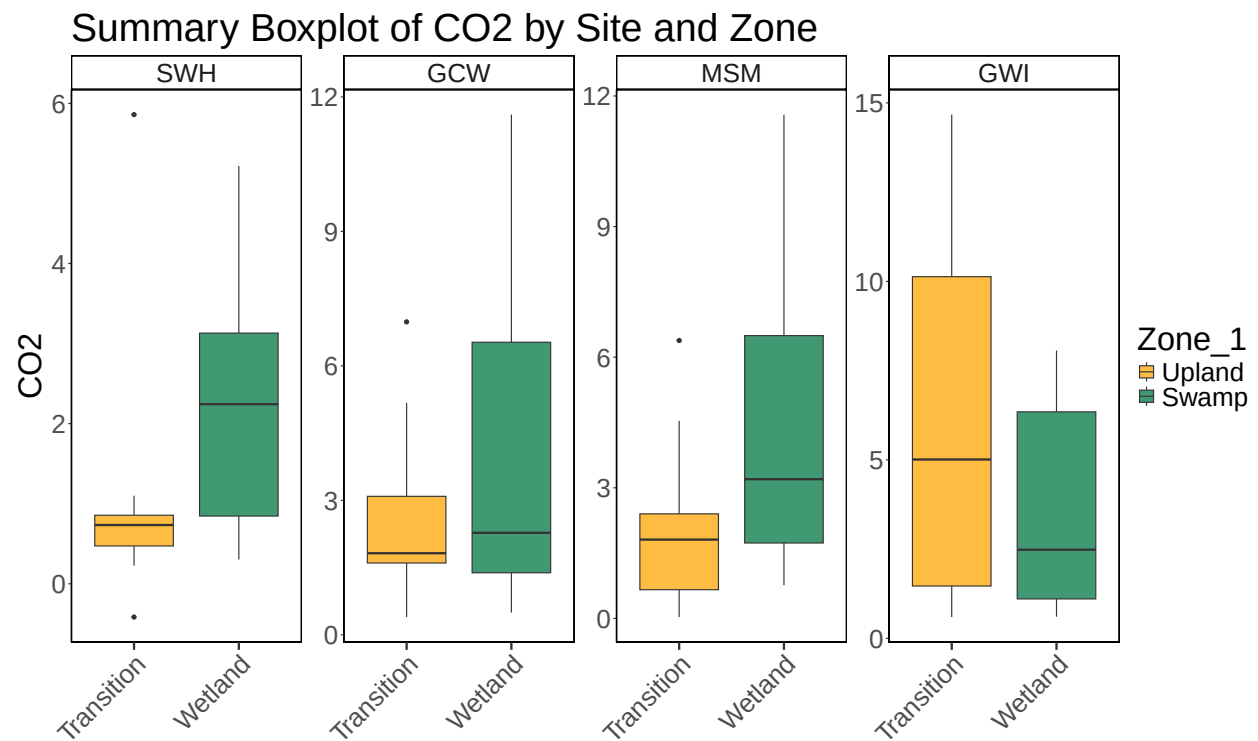
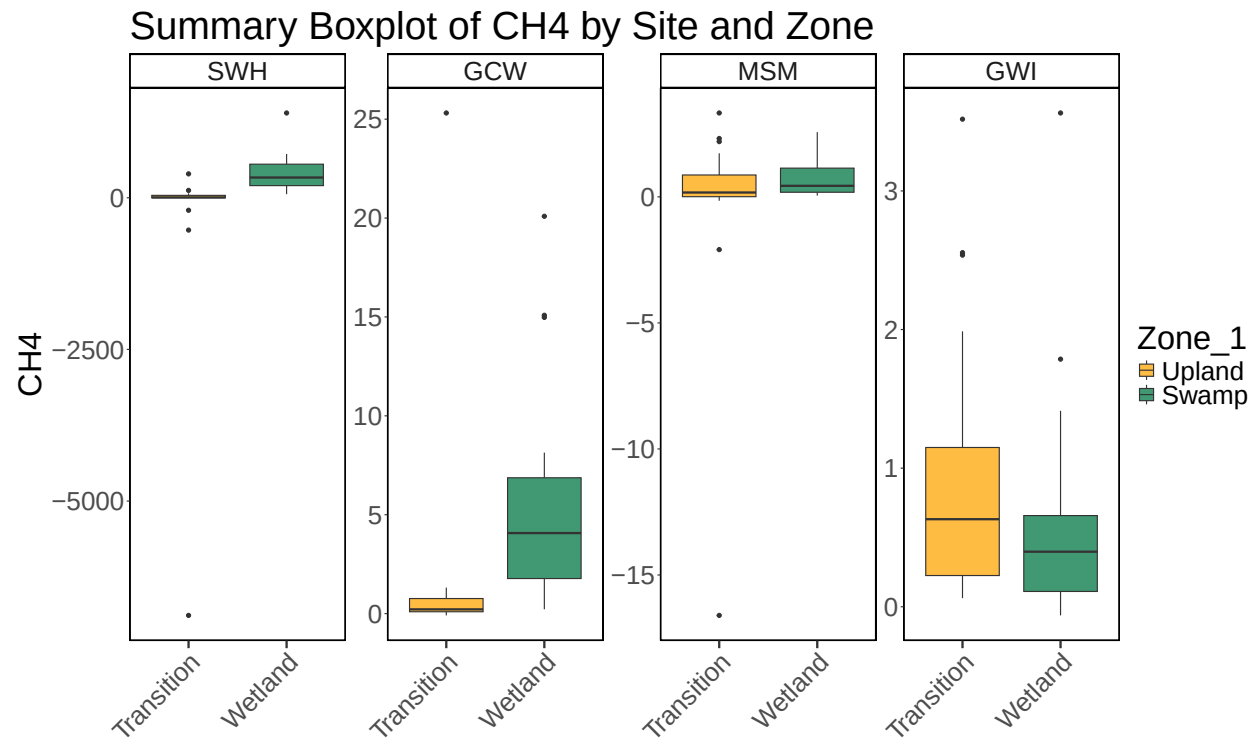
0.2 All Data by Year, Site and Zone



0.3 Boxplot summary by Date



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Transition	-6884.5280533	394.120415	-460.3864727	-0.4154637	5.859412	0.9607529
SWH	Wetland	59.9871778	1398.120534	406.5684788	0.3028537	5.217695	2.2376064
GCW	Transition	-0.0901019	25.312906	1.5921588	0.3954831	6.980726	2.4825838
GCW	Wetland	0.2215926	20.090639	5.0858040	0.4957703	11.604588	4.1424663
MSM	Transition	-16.6028068	3.327981	-0.2807652	0.0305177	6.382414	1.9209085
MSM	Wetland	0.0514141	2.568636	0.6728730	0.7602931	11.571173	4.0361589
GWI	Transition	0.0621000	3.517843	0.9497686	0.5968420	14.671811	6.1086277
GWI	Wetland	-0.0623885	3.562404	0.5542746	0.6065226	8.062884	3.4680774